AI-Assisted Single-Cell Diagnostic Prototype

Multimodal Optical Microscopy & High-Precision Electrochemical Analysis

CAUTION — RESEARCH PROTOTYPE: *RESEARCH PROTOTYPE NOTICE: This system is developed strictly for research, educational, and computational evaluation purposes. It is not an FDA-cleared medical device and must not be used as a clinical diagnostic tool or to guide medical treatments.*

Sample ID:	UNIT_TEST_SPECIMEN	**Analysis Date:**	2026-10-24 00:47:47
Specimen Type:	Single Erythrocyte (RBC)	**Sensory Setup:**	Optical Brightfield + CV / EIS
Instrumentation:	Potentiostat DAQ (16-bit)	**System Core:**	Multimodal AI Pipeline

Diagnostic Classification Findings

Multimodal Prediction	Confidence	Research Risk Score
HEALTHY	**96.8%**	**0.14 / 1.00**

Multimodal Biomarker Quantitative Extraction

Image Feature	Value
Cell Area	1450.0
Circularity (0-1)	0.91
Aspect Ratio	1.05
Texture Variance	22.0
Mean Intensity	115.0

Electrochemical Feature	Value
Peak Anodic Current (Ipa)	12.3 µA
Charge Transfer Res. (Rct)	1490.0 Ω
Capacitance (Cdl)	2.1 µF
Peak Potential Diff (ΔEp)	76.0 mV
Impedance |Z| at 1 Hz	1540.0 Ω

Analyst Observations

Automated pipeline validation report